1 Properties of the Determinant

Proposition 1.1 (Determinant is multiplicative)**.**

(a) If $S, T \in \mathscr{L}(V)$, then $\det(ST) = (\det S)(\det T)$.

(b) If A and B are $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$.

Exercise 1.2. Prove Proposition 1.1.

Proposition 1.3. *A linear operator $T \in \mathscr{L}(V)$ is an isomorphism if and only if* $\det T \neq 0$. *Furthermore, if T is an isomorphism, then* $\det(T^{-1}) = (\det T)^{-1}$.

Proof. Suppose $\det T \neq 0$. Let v be a nonzero vector in V. Since $\{v\}$ is linearly independent, it can be extended to a basis $\{v, e_2, \ldots, e_n\}$ of V. Let M be a nonzero alternating n-linear form on V (that is, M is not the zero map on V^n). Since $\{v, e_2, \ldots, e_n\}$ is linearly independent and M nonzero, it follows from Proposition 4.6 of worksheet 5.3 that $M(v, e_2, \ldots, e_n) \neq 0$. When then have

$$M(Tv, Te_2, \ldots, Te_n) = (\det T)M(v, e_2, \ldots, e_n) \neq 0.$$

This implies that $Tv \neq 0$, since otherwise we would have $M(Tv, Te_2, \ldots, Te_n) = M(0, Te_2, \ldots, Te_n) = 0$ since M is linear in the first entry. Thus we have proved that if $v \neq 0$ then $Tv \neq 0$. This shows that T is injective, and therefore an isomorphism since V is finite-dimensional. □

Exercise 1.4. Complete the proof of Proposition 1.3 by proving the forward direction.

Corollary 1.5. *The following are equivalent:*

(a) A is invertible.

(b) $\det A \neq 0$.

(c) The homogeneous system $AX = 0$ *has only the unique solution.*

Proof. We've proved previously that (a) $\iff$ (c). An $n \times n$ matrix A is invertible if and only if the operator on $M^{n\times 1}(F)$ whose matrix with respect to the standard basis is A is invertible, thus the previous result shows that A is invertible if and only if $\det A \neq 0$. Hence (a) $\iff$ (b), so all three statements are equivalent. □

Proposition 1.6 (Determinant is a similarity invariant)**.** *Let* $T \in \mathscr{L}(V)$ *and* $S : W \to V$ *be an invertible linear map. Then*

$$\det(S^{-1}TS) = \det T.$$

Proof. Since $S : W \to V$ is invertible, this implies $\dim W = \dim V = n$. Let $M \in W_{\text{alt}}^{(n)}$ and define $M' \in V_{\text{alt}}^{(n)}$ by

$$M'(v_1, \ldots, v_n) := M(S^{-1}v_1, \ldots, S^{-1}v_n)$$

for all $(v_1, \ldots, v_n) \in V^n$. By definition of the determinant of the operator $S^{-1}TS$,

$$M_{S^{-1}TS}(w_1, \ldots, w_n) = \det(S^{-1}TS)(w_1, \ldots, w_n).$$

Exercise 1.7. Complete the proof of Proposition 1.6 by calculating
$M_{S^{-1}TS}(w_1, \ldots, w_n) = M(S^{-1}TSw_1, \ldots, S^{-1}TSw_n)$.

□

Corollary 1.8. *For $T \in \mathscr{L}(V)$, $\det T = \det[T]$ where $[T]$ is the matrix of T with respect to any basis of V.*

Exercise 1.9. Prove Corollary 1.8.

Proposition 1.10 (Determinant of a transpose)**.**

(a) If A is an $n \times n$ matrix then $\det(A^t) = \det(A)$.

(b) If $T \in \mathscr{L}(V)$, then $\det(T^t) = \det(T)$.

Proof.

(a)

$$\begin{aligned}\det(A^t) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)(A^t)_{\sigma(1)1} \cdots (A^t)_{\sigma(n)n} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) A_{1\sigma(1)} \cdots A_{n\sigma(n)}.\end{aligned}$$

The map $A \mapsto \det(A^t)$ is therefore an alternating n-linear form on $M^{n\times 1}(F)$ such that $I \mapsto 1$. Since $A \mapsto \det(A)$ is the unique alternating n-linear form on $M^{n\times 1}(F)$ such that $I \mapsto 1$, we have $\det(A^t) = \det(A)$.

Exercise 1.11. Prove part (b) of Proposition 1.10.

□

2 Evaluating Determinants

We now summarize various results which are helpful for evaluating determinants.

Proposition 2.1. *Let A be an $n \times n$ matrix.*

(a) If either two columns or two rows of A are equal, then $\det(A) = 0$.

(b) If B is obtained from A by swapping either two columns or two rows, then $\det(B) = -\det(A)$.

(c) If B is obtained from A by multiplying either one row or one column by a scalar λ, then $\det(B) = \lambda \det(A)$.

(d) If B is obtained from A by replacing one column by the sum of itself and a scalar multiple of another column, then $\det(B) = \det(A)$.

(e) If B is obtained from A by replacing one row by the sum of itself and a scalar multiple of another row, then $\det(B) = \det(A)$.

Proof. All of these assertions follow from the result that $A \mapsto \det(A)$ and $A \mapsto \det(A^t)$ are both alternating n-linear forms on $M^{n\times 1}(F)$. For example, to prove (d) let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$ and suppose we replace $v_1 \mapsto v_1 + \alpha v_2$ for some $\alpha \in F$ to obtain $B = \begin{bmatrix} v_1 + \alpha v_2 & v_2 & \cdots & v_n \end{bmatrix}$. Then

$$\begin{aligned}
\det(B) &= \begin{vmatrix} v_1 + \alpha v_2 & v_2 & \cdots & v_n \end{vmatrix} \\
&= \begin{vmatrix} v_1 & v_2 & \cdots & v_n \end{vmatrix} + \alpha \begin{vmatrix} v_2 & v_2 & \cdots & v_n \end{vmatrix} \quad \text{(by multilinearity)} \\
&= \begin{vmatrix} v_1 & v_2 & \cdots & v_n \end{vmatrix} + \alpha \cdot 0 \quad \text{(by the alternating property)} \\
&= \begin{vmatrix} v_1 & v_2 & \cdots & v_n \end{vmatrix} \\
&= \det(A).
\end{aligned}$$

□

Exercise 2.2. Prove that replacing the first row of and $n \times n$ matrix A with the sum of itself and a scalar multiple of the second row leaves the determinant unchanged.

Proposition 2.1 leads to a much faster way to evaluate the determinant than direct application formula involving the sum over permutations. Apply Gaussian elimination until you reach an upper-triangular matrix, the determinant of which is the product of the diagonal entries. By Proposition 2.1, this gets multiplied by -1 for each row-interchange and by a scalar for each scaling of a row.

Example 2.3. Consider the matrix $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 3 \end{bmatrix}$. Since A has no zeros, there is no row or column for which calculating $\det(A)$ by cofactor expansion will save us time. Instead, we use the strategy above. We row-reduce A to upper triangular form U by the following sequence of elementary row operations:

$$A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 3 \end{bmatrix} \xrightarrow{R_3 \mapsto R_3 + R_1} \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ 0 & 3 & 5 \end{bmatrix} \xrightarrow{R_2 \mapsto R_2 + 2R_1} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 5 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 5 \\ 0 & 0 & -5 \end{bmatrix} = U.$$

Since U is triangular, $\det(U) = -15$ is the product of the diagonal entries of U. The two row-replacements leave the determinant of A unchanged, while the row-interchange multiplies the determinant of A by -1. Hence

$$\det(A) = -\det(U) = 15.$$

Remark 2.4. Most computers use this method to compute $\det A$: when A is $n \times n$, it can be shown that the computation of $\det A$ by row operations requires only $\mathcal{O}(n^3)$ operations. For $n = 25$, this is only about 15,000 operations, which any modern computer can carry out in a fraction of a second.

Exercise 2.5. Use the method above to calculate the determinant of $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$.

Exercise 2.6. Combine the methods of cofactor expansion and row-reduction to calculate the determinant of $A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$ as efficiently as possible.